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FISHING ROD HOLDING SYSTEM

Abstract

A system may include a base plate. The system may include a plurality of rails attached to the base plate, each of the plurality of rails comprising a plurality of connection points. The system may include at least one rod holder configured to be rotatably coupled to at least one of the plurality of rails at a first connection point. The system may include at least one pivot arm rotatably coupled to the at least one rod holder, the at least one pivot arm configured to be coupled to at least one of the plurality of rails at a second connection point.

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Background/Summary

PRIORITY, CROSS-REFERENCE, AND INCORPORATION [0001] This application claims priority to U.S. Provisional Patent Application No. 63/562,139 filed Mar. 6, 2024. The entire contents of the above are hereby incorporated into this document by reference and made a part of this specification for all purposes, for all that it contains. Moreover, any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet of the present application are hereby incorporated by reference under 37 C.F.R. § 1.57.

BACKGROUND

[0002] Fishing is a popular recreational and commercial activity that is enjoyed throughout the world. There are many types of fishing rods available depending on the type of fishing a consumer might want to enjoy including spinning rods, casting rods, trolling rods, surf rods, fly fishing rods, and ice rods. Furthermore, the length of the fishing rod can affect the casting accuracy, the distance of casting, and the leverage of the rod. These factors often lead to consumers engaging in fishing to bring multiple poles or rods to a fishing outing or activity. However, transporting multiple fishing rods can be troublesome. Fishing rods are often long and fragile which makes them prone to snapping or breaking if not handled carefully. Furthermore, the size and shape of the fishing rods make them cumbersome to carry or to store.

[0003] As such, there exists a need for a device capable of storing multiple fishing rods allowing for easy and safe transportation of fishing rods on a boat or vehicle.

SUMMARY

[0004] In some aspects, the techniques described herein relate to a fishing rod holding system, including a base plate, a plurality of rails attached to the base plate, each of the plurality of rails including a plurality of connection points, and at least one rod holder configured to be rotatably coupled to at least one of the plurality of rails at a first connection point; and at least one pivot arm rotatably coupled to the at least one rod holder, the at least one pivot arm configured to be coupled to at least one of the plurality of rails at a second connection point.

[0005] In some aspects, the techniques described herein relate to a system, wherein the rod holder is configured to be rotated between a first and second angular orientation when attached to the at least one of the plurality of rails.

[0006] In some aspects, the techniques described herein relate to a system, wherein the first angular orientation is an orientation that is substantially parallel to the base plate.

[0007] In some aspects, the techniques described herein relate to a system, wherein the second angular orientation is an orientation that is substantially normal to the base plate.

[0008] In some aspects, the techniques described herein relate to a system, wherein the at least one rod holder is further configured to be detachably coupled to the at least one of the plurality of rails.

[0009] In some aspects, the techniques described herein relate to a system, wherein the at least one pivot arm is further configured to secure the at least one rod holder in an angular orientation when the pivot arm is coupled to the at least one of the plurality of rails.

[0010] In some aspects, the techniques described herein relate to a system, wherein the base plate is configured to be mounted onto an external surface or object.

[0011] In some aspects, the techniques described herein relate to a system, further including a vehicle, wherein the base plate is configured to be secured onto a surface of the vehicle.

[0012] In some aspects, the techniques described herein relate to a system, wherein the pivot arm is detachably coupled to the at least one rod holder.

[0013] In some aspects, the techniques described herein relate to a system, wherein the at least one rod holder is configured to be coupled to the least one of the plurality of rails at the first connection point by inserting a mechanical connector through a rail connection point of the rod holder and the first connection point.

[0014] In some aspects, the techniques described herein relate to a system, wherein the mechanical connectors may include, for example, a screw, a bolt and nut, or a pin.

[0015] In some aspects, the techniques described herein relate to a method for storing fishing rods or other items including providing a fishing rod holding system including: a base plate; a plurality of rails attached to the base plate, each of the plurality of rails including a plurality of connection points; and at least one rod holder configured to be rotatably coupled to at least one of the plurality of rails at a first connection point; and at least one pivot arm rotatably attached to the at least one rod holder, the at least one pivot arm configured to be coupled to at least one of the plurality of rails at a second connection point; and coupling the at least one rod holder to the at least one of the plurality of rails at the first connection point; coupling the pivot arm to the at least one of the plurality of rails at a second connection point; and engaging a fishing rod with the at least one rod holder, thereby securing the fishing rod to the fishing rod holding system.

[0016] In some aspects, the techniques described herein relate to a method, further including altering the angular orientation of the least one rod holder.

[0017] In some aspects, the techniques described herein relate to a method, wherein altering the angular orientation of the at least one rod holder includes detaching the pivot arm from the at least one of the plurality of rails at the second connection point; and reattaching the pivot arm to the at least one of the plurality of rails at a third connection point.

[0018] In some aspects, the techniques described herein relate to a method, wherein altering the angular orientation of the at least one rod holder includes detaching the at least one rod holder from the at least one of the plurality of rails at the first connection point; and reattaching the at least one rod holder to the at least one of the plurality of rails at a third connection point.

[0019] In some aspects, the techniques described herein relate to a method, wherein altering the angular orientation of the at least one rod holder includes changing the angular orientation of the at least one rod holder between 0° degrees and 90°.

[0020] In some aspects, the techniques described herein relate to a method, further including mounting the base plate onto an external surface or object.

[0021] In some aspects, the techniques described herein relate to a method, further including securing the base plate to a surface of a vehicle.

[0022] In some aspects, the techniques described herein relate to a method, wherein coupling the at least one rod holder to the least one of the plurality of rails at the first connection point, includes aligning the first connection point with a rail connection point of the rod holder; and inserting a mechanical connector through the first connection point and the rail connection point.

[0023] In some aspects, the techniques described herein relate to a method, wherein the mechanical connector includes a screw, a bolt and nut, or a pin.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] The detailed description is set forth with reference to the accompanying figures. The use of the same numbers in different figures indicates similar or identical items.

[0025] For this discussion, the devices and systems illustrated in the figures are shown having a multiplicity of components. Various implementations of devices and/or systems, as described herein, may include fewer components, and remain within the scope of the disclosure. Alternatively, other implementations of devices and/or systems may include additional components, or various combinations of the described components, and remain within the scope of the disclosure.

[0026] FIG. 1 shows an isometric view of a fishing rod holding system.

[0027] FIG. 2A shows a front view of a base plate.

[0028] FIG. 2B shows an isometric view of a base plate.

[0029] FIG. 2C shows a front view of a base plate.

[0030] FIG. 3A shows a side isometric view of a telescoping arm.
[0031] FIG. 3B shows a side isometric view of an outer body of the telescoping arm
[0032] FIG. 3C shows a side isometric view of an inner body of the telescoping arm
[0033] FIG. 3D shows a top isometric view of a base connector of the telescoping arm.
[0034] FIG. 3E shows a side isometric view of the base connector of FIG. 3D.
[0035] FIG. 4 shows an isometric view of a rail.
[0036] FIG. 5 shows a partial view of the fishing rod holding system with a plurality of rails secured to a base plate.
[0037] FIG. 6 shows an isometric view of a rod holder.
[0038] FIG. 7 shows a front view of the rod holder.
[0039] FIG. 8A shows an isometric view of a pivot arm.
[0040] FIG. 8B shows a top isometric view of a pivot arm.
[0041] FIG. 9 shows a partial view of a fishing rod holding system with rod holders disposed with different orientations.
[0042] FIG. 10 shows an isometric view of a cart feature.
[0043] FIG. 11A shows an isometric view of a handle feature.
[0044] FIG. 11B shows an isometric view of a bottom portion of the central bar of the handle feature.
[0045] FIG. 11C shows an isometric view of the handle base of the handle feature.
[0046] FIG. 11D shows an isometric view of the bottom portion of the handle base.
[0047] FIG. 11E shows an isometric view of the top portion of the handle base FIG. 12A shows an isometric view of a t-track feature.
[0048] FIG. 12B shows a front view of a t-track feature.

DETAILED DESCRIPTION

[0049] In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. Thus, in some embodiments, part numbers may be used for similar components in multiple figures, or part numbers may vary from figure to figure. The illustrative embodiments described herein are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented here. It will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the Figures, can be arranged, substituted, combined, and designed in a wide variety of different configurations, all of which are explicitly contemplated and make part of this disclosure.

[0050] The following detailed description is directed to certain specific embodiments of the development. Reference in this specification to “one embodiment,” “an embodiment,” or “in some embodiments” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the disclosure. The appearances of the phrases “one embodiment,” “an embodiment,” or “in some embodiments” in various places in the specification are not necessarily all referring to the same embodiment, nor are separate or alternative embodiments necessarily mutually exclusive of other embodiments. Moreover, various features are described which may be exhibited by some embodiments and not by others. Similarly, various requirements are described which may be requirements for some embodiments but may not be requirements for other embodiments. Furthermore, embodiments of the development may include several novel features, no single one of which is solely responsible for its desirable attributes or which is essential to practicing any particular embodiment described herein.

[0051] FIG. 1 shows an isometric view of a fishing rod holding system **100**. As seen in FIG. 1, the system **100** includes a base plate **120**, a plurality of rails **140**, a plurality of rod holders **160**, and a plurality of pivot arms **180**. The rails **140** are configured to be secured to the base plate **120** to create a plurality of connection channels **150** between adjacent rails **140**. The pivot arms **180** and

rod holders **160** are configured to be secured to the rails **140** with the connection channels **150** in a variety of configurations as described herein.

[0052] FIG. 2A shows a front view of a base plate **120**. As seen in FIG. 2A, the base plate **120** may include a plurality of connection points, such as a plurality of rail connection points **122**, a plurality of large connection points **124**, and a plurality of mounting connection points **128**. In some embodiments, the base plate **120** includes a plurality of slots **126**.

[0053] The base plate **120** may be substantially rectangular in shape. In some embodiments, the base plate may take a different shape including but not limited to a circular shape, an oval shape, a square shape, a hexagon shape, or any other polygon shape.

[0054] In some embodiments, the base plate **120** is formed from any suitable material, such as a polymer or plastic material, (e.g., polyethylene, high density polyethylene, low density polyethylene, polypropylene, polyvinyl chloride, polycarbonate, resin etc.), a metal material or alloy (e.g., stainless steel, titanium, aluminum, iron, copper, brass, bronze, lead, nickel, etc.) or a composite material (e.g., fiber glass, carbon fiber, etc.). In some embodiments, the base plate **120** is formed of polyethylene. In some embodiments, the base plate **120** is formed of high density polyethylene. In some embodiments, the base plate **120** is formed via injection molding.

[0055] The plurality of rail connection points **122** are disposed on a front surface of the base plate **120**. In some embodiments, the rail connection points **122** may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole a tapered hole, a keyhole slot, a dowel hole, or a helical hole. In some embodiments, the rail connection points penetrate from one side of the base plate **120** to emerge on an opposite side of the base plate **120**. In some embodiments, the rail connection points **122** do not penetrate completely through the base plate **120**.

[0056] In some embodiments, the plurality of rail connection points **122** are disposed on the front surface of the base plate **120** in a plurality of columns **123**. In some embodiments, a column of rail connection points **123** may be used to assist in securing a rail **140** to the base plate **120**. In some embodiments, a column of rail connection points **123** may include 1, 2, 3, 4, or 5 rail connections points **122**. In some embodiments, more than 5 rail connections points **122** may be included in a column **123**.

[0057] In some embodiments, the plurality of large connection points **124** is disposed on the front surface of the base plate **120**. The plurality of large connection points **124** may consist of holes that penetrate the through the base plate **120**. In some embodiments, each of the large connection points **124** may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole a tapered hole, a keyhole slot, a dowel hole, or a helical hole. In some embodiments, the large connection points **124** do not penetrate completely through the base plate **120**. The large connection points **124** may serve to secure the base plate **120** to an external surface or object, such as a wall, a vehicle, such as a truck or boat, or other area where a user may desire to mount or secure the base plate **120**. In some embodiments, a vehicle may include an automobile, such as a car, truck, bus, van, SUV, etc., a motorcycle, an off road vehicle, such as an ATV, a scooter, a bicycle, a train, a ship, a boat, such as commercial fishing boats, a bass boat, a skiff, a drift boat, a Jon boat, a yacht, or other types of watercraft, or a personal watercraft, such as jet skis, etc.

[0058] In some embodiments, the base plate **120** is secured to an external object or surface by aligning the large connection points **124** with an external surface or object and securing the base plate **120** by means of a mechanical connector disposed through the large connection point **124** and the corresponding feature of the external surface or object. In some embodiments, an external feature such as the telescoping arm, cart feature, and t-track feature as described below are secured to the base plate **120** by aligning at least one large connection points **124** with a corresponding feature on the external feature and securing the external feature to the base plate **120** by means of a mechanical connector disposed through the large connection point **124** and the corresponding

feature of the external feature. In some embodiments, mechanical connectors may include, but are not limited to, bolts and nuts, dowels, screws, clips, clamps, fasteners, pins, or keys. In some embodiments, the mechanical connector may be inserted or threaded through a large connection point **124** and through a corresponding hole or connection point of an external object or surface or external feature or through the external object or surface or external feature itself, thereby securing the base plate to the external surface or object or external feature. In some embodiments, the large connection points **124** may serve to connect an external feature to the base plate **120**.

[0059] The plurality of slots **126** may each be largely rectangular in shape and disposed along an outer perimeter of the base plate **120**. In some embodiments, the base plate **120** may have slots **126** of different sizes and shapes. In some embodiments, the slots **126** may take a different shape including but not limited to a circular shape, an oval shape, a square shape, a hexagon shape, or any other polygon shape. In some embodiments, the slots **126** serve to hang or secure the base plate on an external object or surface. In some embodiments, the slots **126** allow the base plate to be hung on external hooks or knobs or other object. In some embodiments, the slots **126** serve as handholds for a user, thereby allowing a user to easily carry the base plate **120**.

[0060] In some embodiments, the base plate **120** also includes a plurality of mounting points **128**. In some embodiments, the plurality of mounting points **128** may be disposed adjacent to at least a portion of the slots **126**. In some embodiments, the mounting points **128** may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole a tapered hole, a keyhole slot, a dowel hole, or a helical hole. In some embodiments, the mounting points **128** may serve to attach a hook, a loop or other mechanical device. The hook, loop, D-ring, or other mechanical device may serve as an additional means for mounting or hanging the base plate **120**. In some embodiments, the large mounting points **128** may serve to secure an external feature such as the telescoping arm, cart feature, and t-track feature as described below to the base plate **120**. In some embodiments, an external feature such as the telescoping arm, cart feature, and t-track feature as described below are secured to the base plate **120** by aligning at least one mounting point **128** with a corresponding feature on the external feature and securing the external feature to the base plate **120** by means of a mechanical connector being disposed through the mounting point **128** and the corresponding feature of the external feature.

[0061] FIG. 2B shows an isometric view of a base plate **120**. As seen in FIG. 2B, the base plate **2B** may vary in size. Additionally, as seen in FIG. 2B, the plurality of large connection points **124** may be disposed on the base plate **120** in a plurality of columns **125**. Each column of large connection points **125** may be disposed between adjacent columns of rail connection points **123**. This configuration beneficially allows for greater flexibility and choice by a user when attaching external features to the base plate **120**.

[0062] FIG. 2C shows a front view of the base plate **120**. As seen in FIG. 2C, in some embodiments, the base plate **120** may include friction points **121**. The friction points **121** may be disposed on a perimeter of the base plate **120** and be configured to receive a friction device such as rubber feet or other similar device. The friction device may be disposed on an underside of the base plate **120** such that the friction device contacts the surface the base plate **120** is disposed on. In some embodiments, the friction points may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole a tapered hole, a keyhole slot, a dowel hole, or a helical hole.

[0063] Additionally, as seen in FIG. 2C, the base plate **120** may also include at least one first arm connection point **127** and at least one second connection point **129**. In some embodiments, the first arm connection points **127** may be disposed near or on the perimeter of the base plate **120** and be configured to assist in securing a telescoping arm (such as the telescoping arm **300** described below). The at least one second connection point **129** may be disposed around the at least one arm connection point **127**. In some embodiments, each first arm connection point **127** may be surrounded by three second arm connection points **129**. In some embodiments, the second arm

connections points **129** may be disposed at 90-degree intervals around the first arm connection point **127**. Thereby allowing greater flexibility for securing the telescoping arm **300** to the base plate **120**.

[0064] In some embodiments, at least one telescoping arm **300** may be attached to the base plate **120** at or near the location of the plurality of mounting points **128**. In some embodiments, the at least one telescoping arm is connected to base plate via at least one large connection points **124**. FIG. **3A** shows an isometric view of a telescoping arm **300**. As seen in FIG. **3A**, the telescoping arm **300** may have an outer body **310**, an inner body **320**, a base connector **350** and an external connector **340**.

[0065] FIG. **3B** shows an isometric view of the outer body **310**. The outer body **310** may be substantially cylindrical in shape with a hollow center. In some embodiments, the outer body **310** have a cross-section that is substantially square or rectangular in shape with a hollow center. At a first end **312** of the outer body **310**, the outer body **310** may include at least one lower connection point **314** and at least one upper connection point **316**. In some embodiments, the outer body **310** may be rounded at the first end **312**. In some embodiments, there are two lower connection points **314** and two upper connection points **316**, a lower connection point **314** and an upper connection point **316** disposed on a first side **311** of the outer body **310** and a lower connection point **314** and an upper connection **316** disposed on a second side **313** of the outer body **310**. In some embodiments, the lower connection points **314** are aligned along a length of outer body **310** such that a mechanical connector can be disposed through each lower connection point **314** simultaneously. In some embodiments, the upper connection points **316** are aligned along a length of outer body **310** such that a mechanical connector can be disposed through each upper connection point **316** simultaneously. In some embodiments, each of the lower connection points **314** and the upper connection points **316** may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole, a tapered hole, a keyhole slot, a dowel hole, or a helical hole.

[0066] At a second end **317** of the outer body **310**, the outer body **310** may be substantially hollow, thereby allowing the inner body **320** to be inserted into the outer body **310** of the telescoping arm **300**. In some embodiments, the outer body **310** may be hollow along the entire length of the outer body **310**. The second end **317** of the outer body **310**, may also include at least one outer extension hole **318**. In some embodiments, there are two outer extension points **318**, an outer extension point **318** on the first side **311** of the outer body **310** and an outer extension point **318** on the second side **313** of the outer body **310**. In some embodiments, the outer extension points **318** are aligned along a length of outer body **310** such that a mechanical connector can be disposed through each outer extension point **318** simultaneously. In some embodiments, each of the outer extension points **318** may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole a tapered hole, a keyhole slot, a dowel hole, or a helical hole.

[0067] FIG. **3C** shows an isometric view of the inner body **320** with the external connector **340**. In some embodiments, the inner body **320** may be substantially cylindrical in shape. The inner body **320** may have a diameter that is smaller than the diameter of the outer body **310** such that the inner body **320** can be inserted into the outer body **310**. In some embodiments, the inner body **320** may have a cross-section that is substantially square or rectangular in shape. In some embodiments, the cross-section of the inner body **320** may be generally square or rectangular in shape, while missing a side, thereby forming an open rectangle and forming a general C-shape. The cross section of the inner body **320** may be smaller than the cross section of outer body **310** such that the inner body **320** can be inserted into the hollow center of the outer body **310**. The inner body **320** includes a plurality of inner extension holes **322**. The plurality of inner extension holes **322** may be disposed along the length of the inner body **320**. In some embodiments, each inner extension hole **322** may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole, a tapered hole, a keyhole slot, a dowel hole, or a helical hole.

[0068] The overall length of the telescoping arm **300** can be varied by inserting the inner body **320** of the telescoping arm **300** into the outer body **310** to a desired depth. The telescoping arm **300** can be set to a desired length by inserting the inner body **320** of the telescoping arm **300** into the outer body **310**, aligning an inner extension hole **322** of the inner body **320** with a set of outer extension points **318** of the outer body **310**, and securing by means of a mechanical connector being disposed through the aligned inner extension hole **322** of the inner body **320** and outer extension points **318** of the outer body **310**. In some embodiments, mechanical connectors may include, but are not limited to, bolts and nuts, dowels, screws, clips, clamps, fasteners, pins, or keys. In some embodiments, the mechanical connector may be disposed or threaded through a set of outer extension points **318** aligned with an inner extension hole **322**, thereby securing the inner body **320** within the outer body **310**.

[0069] The inner body **320** is connected to the external connector **340**. In some embodiments, the external connector **340** is integrally formed with the inner body **320**. In some embodiments, the external connector **340** is formed separately from the inner body **320** and is configured to be inserted and/or secured to the inner body **320**. This configuration may be beneficial for allowing a user to change the external connector **340** to a desired connection mechanism. In some embodiments, the external connector **340** may include, but is not limited to, a snap hook, a carabiner, s-hook, spring clip, clip, quick link, or snap ring. In some embodiments, the external connector **340** includes a first prong **342** and a second prong **344**. The first and second prongs **342**, **344** may be largely rectangular in shape and disposed parallel to each other. The first and second prongs **342**, **344** may be separated by a gap **346**. The first prong **342** may have a first external connection point **345** disposed on an upper end **347** of the first prong **342**. The second prong **344** may have a second external connection point **348** disposed on an upper end **349** of the second prong **344**. In some embodiments, the first and second external connection points **345**, **348** are aligned along the lengths of the first and second prongs **342**, **344** such that a mechanical connector can be disposed through each of the first and second external connection points **345**, **348** simultaneously. In some embodiments, each of the first and second external connection points **345**, **348** may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole, a tapered hole, a keyhole slot, a dowel hole, or a helical hole.

[0070] The external connector **340** is configured to engage with the connection mechanism on an external surface, such as within a boat, truck, or other vehicle, some of which are listed herein. In some embodiments, a feature of the external surface or object or vehicle is inserted into the gap **346** of the external connector **340**, where a protrusion or hole of the external surface or vehicle may be aligned with the first and second external connection points **345**, **348** of the external connector **340**. A mechanical connector may then be disposed or threaded through the protrusion or hole of the external surface or object and the first and second external connection points **345**, **348** of the external connector **340**, thereby securing the telescoping arm to the external surface or object or vehicle. The external connector **340** is configured to secure the telescoping arm **300** and thereby the base plate **120** and the system **100** to the external surface or vehicle and to prevent the system **100** from moving or tilting while connected to the external surface.

[0071] FIG. 3D shows an upper isometric view of the base connector **350**. As seen in FIG. 3D, the base connector **350** may include a connector base **352**, a first wall **358** and a second wall **360**. The connector base **352** may be substantially flat and in a rectangular shape, with the exception of a protrusion **353** integrally formed with the connector base **352**. In some embodiments, the protrusion **353** may be generally oval in shape. The connector base **352** may include a first base connection point **354** and a plurality of second base connection points **356**. As seen in FIG. 3D, the first base connection point **354** is disposed on the connector base **352** and may be centrally located on the connector base **352**.

[0072] The first base connection point **354** is configured to secure the base connector **350** to a base

plate **120**. In some embodiments, the first base connection point **354** may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole, a tapered hole, a keyhole slot, a dowel hole, or a helical hole. In some embodiments, the first base connection point **354** is configured to align with a first arm connection point **127** of the base plate **120** or other hole or connection point, such as a large connection point **124** or mounting point **128**, on the base plate **120** and a mechanical connector, such as those described herein, may be threaded through the first base connection point **354** and the mounting point **128** or large cone connection point **124** on the base plate **120** to secure the base connector **350** to the base plate **120**. In some embodiments, when the base connector **350** is secured to the base plate **120** via the first base connection point **354**, the base connector **350** is able to rotate along an axis extending normal to the connector base **352**.

[0073] The plurality of second base connection points **356** are disposed on the protrusion **353**. In some embodiments, the plurality of second base connection points **356** are disposed on a perimeter of the protrusion **353**. In some embodiments, each second base connection points **356** of the plurality of second base connection points are equidistant from the first base connection point **354**. In some embodiments, each second base connection point **356** of the plurality of second base connection points **356** may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole, a tapered hole, a keyhole slot, a dowel hole, or a helical hole. In some embodiments, a second base connection point **356** is configured to align with a second arm connection point **129**, mounting point **128**, or large connection point **124** of the base plate **120** and a mechanical connector, such as those described herein, may be threaded through the aligned second base connection point **356** and the second arm connection point **129**, mounting point **128**, or large connection point **124** on the base plate **120** to further secure the base connector **350** to the base plate **120**. In some embodiments, when the base connector **350** is secured to the base plate **120** via the first base connection point **354**, the base connector **350** is able to rotate on an axis extending normal to the connector base **352**. The base connector **350** can be secured in a desired orientation by rotating the base connector **350** to the desired orientation and then securing the base connector **350** to the base plate **120** by aligning a second base connection point **356** with a second arm connection point **129**, mounting point **128**, or large connection point **124** and disposing a mechanical connector through the aligned points. In some embodiments, the plurality of second base connection points **356** allow a change in orientation up to 360 degrees. In some embodiments, the plurality of second base connection points **356** allow a change in orientation up to 180 degrees, up to 90 degrees up to 60 degrees, up to 45 degrees, or up to 30 degrees.

[0074] FIG. 3E shows a side isometric view of the base connector **350**. As seen in FIG. 3E, the first and second walls **358**, **360** extend in a direction normal to the connector base **352**. In some embodiments, the first and second walls **358**, **360** may be connected to the connector base **352** such that a gap **362** is formed in between the first and second walls **358**, **360**. In some embodiments, the first and second walls **358**, **360** may be generally semi-circular in shape. In some embodiments, the first and second walls **358**, **360** may be generally rectangular in shape.

[0075] As seen in FIG. 3E, the first and second walls **358**, **360** may each include a central connection point **364** and a plurality of perimeter connection points **366**. The central connection point **364** may be disposed centrally on each of the first and second walls **358**, **360**. In some embodiments, the central connection point **364** on each of the first and second walls **358**, **360** are aligned such that a mechanical connector can be disposed through each central connection point **364** simultaneously. In some embodiments, each central connection point **364** may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole, a tapered hole, a keyhole slot, a dowel hole, or a helical hole.

[0076] The plurality of perimeter connection points **366** are disposed on a perimeter of each of the first and second walls **358**, **360**. In some embodiments, each connection point **366** of the plurality

of perimeter connection points **366** on each of the first and second walls **358**, **360** are aligned such that a mechanical connector can be disposed through aligned perimeter connection points **366** on each of the first and second walls **358**, **360** simultaneously. In some embodiments, each perimeter connection point **366** may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole, a tapered hole, a keyhole slot, a dowel hole, or a helical hole.

[0077] The base connector **350** may be configured to receive the outer body **310** in the gap **362**. The outer body **310** can then be secured to the base connector **350** by aligning the lower connection points **314** of the outer body **310** with the central connection points **364** of the base connector **350**. A mechanical connector may then be disposed through the aligned central connection points **364** and the lower connection points **314** thereby securing the outer body **310** to the base connector **350**. When the outer body **310** is secured to the base connector **350** by the central connection points **364** and the lower connection points **314**, the outer body **310** may be free to rotate about an axis extending normally from the first and second walls **358**, **360**. A user may then rotate the outer body **310** to a desired orientation. Once the outer body **310** is in a desired orientation, the outer body **310** can be secured in the desired orientation by disposing a mechanical connector through corresponding perimeter connection points **366** of the base connector **350** and the upper connection points **316** of the outer body **320**. This beneficially allows a user to change the orientation of the telescoping arm **300**. In some embodiments, the orientation of the telescoping arm **300** may be changed up to 180 degrees.

[0078] The telescoping arm **300** is configured to be secured to the base plate **120** via the base connector **350**. In some embodiments, the telescoping arm **300** is configured to be secured to the base plate **120** in a variety of configurations, such that the telescoping arm **300** can be adjusted in both the horizontal and vertical planes. These configurations are beneficial for allowing the telescoping arm **300** to be secured to a variety of external surfaces, even if the external surfaces have inconsistent connection points.

[0079] In some embodiments, the telescoping arm **300** may be configured to be folded against the base plate **120**. When the telescoping arm **300** is in this configuration, the telescoping arm **300** may serve as an anchoring point for a strap, hook, clip, or other device for securing the base plate **120** to an external surface.

[0080] FIG. **4** shows an isometric view of a rail **140**. As seen in FIG. **4**, the rail **140** may be largely rectangular in shape. In some embodiments, the length of the rail **140** may be less than a width of the base plate **120**. This configuration may be beneficial for attaching the rail **140** to the base plate **120** without a portion of the rail **140** protruding out from a side of the base plate **120**. In some embodiments, the length of the rail **140** may be equal to or greater than a width of the base plate **120**.

[0081] Each rail **140** is formed from any suitable material, such as a polymer or plastic material, (e.g., polyethylene, high density polyethylene, low density polyethylene, polypropylene, polyvinyl chloride, polycarbonate, resin etc.), a metal material or alloy (e.g., stainless steel, titanium, aluminum, iron, copper, brass, bronze, lead, nickel, etc.) or a composite material (e.g., fiber glass, carbon fiber, etc.). In some embodiments, each rail **140** is formed of polyethylene. In some embodiments, each rail **140** is formed of high density polyethylene. In some embodiments, each rail **140** is formed via injection molding.

[0082] The rail **140** includes a plurality of plate connection points **142** and a plurality of holder connection points **144**. The plate connection points **142** may be disposed on at least one side of the rail **140**. In some embodiments, the plate connection points **142** are disposed on a first surface **148** of the rail **140**. In some embodiments, each rail **140** may have one, two or three plate connection points. In some embodiments, each rail **140** has more than three plate connection points **142**, such as, but not limited to, four, five, or six plate connection points **142**. In some embodiments, the distance between adjacent plate connection points **142** may be the same along the length of the rail

140.

[0083] In some embodiments, the plate connection points **142** may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole a tapered hole, a keyhole slot, a dowel hole, or a helical hole. In some embodiments, the plate connection points **142** penetrate through one side of the rail **140** to emerge on an opposite side of the rail **140**. In some embodiments, the plate connection points **142** do not penetrate completely through the rail **140**.

[0084] The rail **140** may be secured to the base plate **120** by aligning the plate connection points **142** with at least a portion of the rail connection points **122** of the base plate **120** and securing the rail **140** to the base plate **120** by means of one of more mechanical connectors being disposed through the aligned rail connection points **122** and plate connection points **142**. In some embodiments, mechanical connectors may include, but are not limited to, bolts and nuts, dowels, screws, clips, clamps, fasteners, pins, or keys. In some embodiments, the mechanical connector may be inserted or threaded through a plate connection point **142** and through the rail connection point **122**, thereby securing the rail **140** to the base plate **120**.

[0085] The rail **140** further includes a plurality of holder connection points **144**. The holder connection points **144** may be disposed on at least one side of the rail **140**. In some embodiments, the holder connection points **144** are disposed on a different side of the rail **140** as the plate connection points **142**. In some embodiments, the holder connection points **144** are disposed on a second surface **152** of the rail. In some embodiments, the second surface **152** is orthogonal to the first surface **148**. In some embodiments, the holder connection points **144** are disposed on the first surface **148** of the rail **140**, in between or adjacent to the plate connection points **142**. In some embodiments, each rail **140** may have any number of holder connection points **144**, including at least about 1 and/or equal to or less than 50 holder connection points **144**, at least about 5 and/or less than or equal to 25 holder connections points **144**, or at least about 10 and/or less than or equal to about 20 holder connection points **144**. In some embodiments, the distance between adjacent holder connection points **144** is the same along the length of the rail **140**.

[0086] In some embodiments, each holder connection point **144** may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole a tapered hole, a keyhole slot, a dowel hole, or a helical hole. In some embodiments, at least of the portion of the holder connection points **144** penetrate completely through one side of the rail **140** to emerge on an opposite side of the rail **140**. In some embodiments, the holder connection points **144** do not penetrate completely through the rail **140**.

[0087] In some embodiments, the rail **140** includes a plurality of indents **146**. In some embodiments, the indents **146** are located on the same sides of the rail **140** as the holder connection points **144**. Each indent **146** may be located in between adjacent holder connection points **144**. In some embodiments, each indent **146** may be substantially shaped like an hourglass. In some embodiments, the indents **146** may have a different shape, including, but not limited to, a circle shape, an oval, a square, a rectangle, a hexagon, or any other polygon. In some embodiments, the indents **146** may be beneficial for reducing the amount of material used for the rail **140**, thereby reducing the cost and weight of each rail **140**.

[0088] FIG. 5 shows a partial view of the system **100** with a plurality of rails **140** secured to a base plate **120**. As seen in FIG. 5, a plurality of rails **140** may be secured to the base plate **120** by aligning the rail connection points **122** of the base plate **120** with the plate connection points **142** of the rails **140** and securing them by means of at least one mechanical connector being disposed through the aligned plate connection points **142** and rail connection points **122**. In some embodiments, the mechanical connector may be inserted or threaded through a plate connection point **142** and through the rail connection point **122**, thereby securing the rail **140** to the base plate **120**. In some embodiments, the distance between each adjacent rail **140** may be substantially the same along the length of the base plate **120**. In some embodiments, a connection channel **150** may

be formed between each pair of adjacent rails **140** secured to the base plate **120**. The rod holders **160** and pivot arms **180** may be connected to the rail **140** in the connection channels **150** as described herein.

[0089] FIG. **6** shows an isometric view of a rod holder **160**. As seen in FIG. **6**, the rod holder **160** may include a body **162**, a central orifice **164**, a rail connector **170**, a pivot connector **174**, and at least one hook **176**.

[0090] In some embodiments, the rod holder **160** is formed from any suitable material, such as a polymer or plastic material, (e.g., polyethylene, high density polyethylene, low density polyethylene, polypropylene, polyvinyl chloride, polycarbonate, resin etc.), a metal material or alloy (e.g., stainless steel, titanium, aluminum, iron, copper, brass, bronze, lead, nickel, etc.) or a composite material (e.g., fiber glass, carbon fiber, etc.). In some embodiments, the rod holder **160** is formed of polyethylene. In some embodiments, the rod holder **160** is formed of high density polyethylene. In some embodiments, the rod holder **160** is formed via injection molding.

[0091] The body **162** of the rod holder **160** may be substantially cylindrical in shape. The body **162** may be substantially hollow and form the boundaries of an orifice **164**. The orifice **164** is formed from the body **162**. In some embodiments, the orifice **164** only extends a certain distance through the body **162**. In some embodiments, the orifice **164** extends through the entire body **162**, thereby forming an opening on opposite sides of the rod holder **160**. The orifice **164** is configured to receive the handle of a fishing rod, thereby securing the rod within the rod holder **160**. In some embodiments, the orifice **164** is configured to receive and secure other cylindrical shaped objects.

[0092] The rail connector **170** is formed on a first end of the rod holder **160**. In some embodiments, the rail connector **170** is located on a side opposite to the opening of the orifice **164**. In some embodiments, the rail connector **170** is integrally formed with the body **162** of the rod holder **160**. The rail connector **170** may include a plurality of wings **172**. Each wing **172** may extend laterally from the body **162** of the rod holder **160**. The distance between one end of a wing **172** to the end of another wing **172** may be equal to the width of a connection channel **150** as described herein.

[0093] Each wing **172** may include a rail connection point **173**. In some embodiments, the rail connection point **173** may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole a tapered hole, a keyhole slot, a dowel hole, or a helical hole. In some embodiments, the rail connection point **173** penetrates through one side of the wing **172** to emerge on an opposite side of the wing **172**. In some embodiments, the rail connection point **173** does not penetrate completely through the wing **172**.

[0094] The rod holder **160** may be secured to a rail **140** or a pair of adjacent rails **140** by aligning the rail connection points **173** of the rod holder **160** with holder connection points **144** of adjacent rails **140** and securing by means of a mechanical connector. In some embodiments, mechanical connectors may include, but are not limited to, bolts and nuts, dowels, screws, clips, clamps, fasteners, pins, or keys. In some embodiments, the mechanical connector may be inserted or threaded through a rail connection point **173** and through the holder connection point **144**, thereby securing the rod holder **160** to the rail **140**.

[0095] In some embodiments, the rail connector **170** includes a surface **177** that further connects the wing **172** to the body **162** of the rod holder **160**. The surface **177** may be triangular in shape and extend from an end of a wing **172** to a center point of the body **162** of the rod holder **160**. In some embodiments, the surface **177** may be thinner than the rail connector **170** such that a lip **178** is formed from the connection of the surface **177** to the rail connector **170** and/or wing **172**.

[0096] The rod holder **160** further includes a pivot connector **174**. The pivot connector **174** may be centrally located along the length of the rod holder **160**. The pivot connector **174** may extend from the rod holder **160** such that it is largely parallel to the wings **172** of the rail connector **170**.

[0097] The pivot connector **174** may include a plurality of pivot connection points **175**. In some embodiments, the pivot connector **174** includes two pivot connection points **175**. Each pivot connection point **175** may consist of a hole, such as, but not limited to, a round hole, a countersunk

hole, a counterbore hole, a through hole, a blind hole a tapered hole, a keyhole slot, a dowel hole, or a helical hole. In some embodiments, the pivot connection points **175** penetrates through one side of the pivot connector **174** to emerge on an opposite side of the pivot connector **174**. In some embodiments, the pivot connection points **175** do not penetrate completely through the pivot connector **174**.

[0098] The rod holder **160** includes at least one hook **176**. The hook or hooks may be disposed along the length of the body **162** of the rod holder **160** in such a manner that the hook **176** is substantially perpendicular to the pivot connector **174** or rail connector **170**. The hook **176** may be useful for storing the rod holder **160** when not in secured to a rail **140**.

[0099] FIG. **7** shows a front view of the rod holder **160**. As seen in FIG. **7**, the rod holder **160** may include at least one hole **166** and a support **168**. The plurality of holes **166** may be formed on an end of the body **162** of the rod holder **160** opposite to the opening of the orifice **164**. In some embodiments, the at least one hole **166** may include 1, 2, 3, 4, 5, 6, 7, 8 or more holes. The holes **166** may serve to prevent a vacuum from being formed with the orifice **164** or prevent air from being trapped in the orifice **164** when a rod holder or other cylindrical object is inserted in the rod holder **160**.

[0100] The supports **168** may be a raised surface formed on the same surface as the holes **166**. The support **168** may be substantially cross-shaped in nature. The support **168** may serve to support a fishing rod or other cylindrical object when disposed within the rod holder **160**.

[0101] FIG. **8A** shows an isometric view of pivot arm **180**. In some embodiments, the pivot arm **180** is roughly triangular in shape. In some embodiments, the pivot arm **180** is formed from any suitable material, such as a polymer or plastic material, (e.g., polyethylene, high density polyethylene, low density polyethylene, polypropylene, polyvinyl chloride, polycarbonate, resin etc.), a metal material or alloy (e.g., stainless steel, titanium, aluminum, iron, copper, brass, bronze, lead, nickel, etc.) or a composite material (e.g., fiber glass, carbon fiber, etc.). In some embodiments, the pivot arm **180** is formed of polyethylene. In some embodiments, the pivot arm **180** is formed of high density polyethylene. In some embodiments, the pivot arm **180** is formed via injection molding.

[0102] The pivot arm **180** may include a top portion **182** and a bottom portion **184**. The top portion **182** may include a plurality of holder connectors **186**. In some embodiments, the pivot arm **180** may include two holder connectors **186**. The holder connectors **186** may be spaced such that there is a gap **188** between the holder connectors **186**. The gap **188** may be of a sufficient width that the pivot connector **174** of the rod holder **160** is able to be inserted into the gap **188**.

[0103] Each holder connector **186** may include at least one holder connection point **187a**, **187b**. Each holder connection point **187a**, **187b** may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole a tapered hole, a keyhole slot, a dowel hole, or a helical hole. In some embodiments, the holder connection point **187a**, **187b** penetrates through one side of the holder connector **186** to emerge on an opposite side of the holder connector **186**. In some embodiments, the holder connection points **187a**, **187b** do not penetrate completely through the holder connector **186**.

[0104] The pivot arm **180** may be secured to the rod holder **160** by inserting the pivot connector **174** of the rod holder **160** between the holder connectors **186** of the pivot arm **180**, aligning at least one of the pivot connections points **175** with at least one of the holder connection points **187a**, **187b** of each of the holder connectors **186**, and securing by means of a mechanical connector. In some embodiments, mechanical connectors may include, but are not limited to, bolts and nuts, dowels, screws, clips, clamps, fasteners, pins, or keys. In some embodiments, the mechanical connector may be inserted or threaded through a first holder connection point **187a**, through a pivot connection point **175**, and through the second holder connection **187b** point thereby securing the pivot arm **180** to the rod holder **160**.

[0105] The bottom portion **184** of the pivot arm **180** includes at least one wing **190a**, **190b**. The

distance between an end of a first wing **190a** and an end of a second wing **190b** may be equal to or substantially similar to the width of a connection channel **150**. Each wing **190a**, **190b** may include at least one rail connection point **191**. In some embodiments, each rail connection point **191** may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole, a tapered hole, a keyhole slot, a dowel hole, or a helical hole. In some embodiments, the rail connection point **191** penetrates through one side of the wing **190a**, **190b** to emerge on an opposite side of the wing **190a**, **190b**. In some embodiments, the rail connection point **191** does not penetrate completely through the wing **190a**, **190b**.

[0106] The pivot arm **180** may be secured to a rail **140** or a pair of adjacent rails **140** by aligning the rail connection points **191** of the pivot arm **180** with holder connection points **144** of adjacent rails **140** and securing by means of a mechanical connector. In some embodiments, mechanical connectors may include, but are not limited to, bolts and nuts, dowels, screws, clips, clamps, fasteners, pins, or keys. In some embodiments, the mechanical connector may be inserted or threaded through a rail connection point **191** and through the holder connection point, thereby securing the pivot arm **180** to the base rail.

[0107] FIG. **8B** shows a top isometric view of the pivot arm **180**. As seen in FIG. **8B**, in some embodiments, the body **193** of the pivot arm **180** may be substantially hollow thereby forming an arm channel **195** within the body **193**. In some embodiments, the channel **195** may include a plurality of arms **197** connecting a first side **196** of the arm channel **195** to a second side **198** of the arm channel. This configuration may be beneficial for accommodating injection molding manufacturing of the pivot arm **180**.

[0108] FIG. **9** shows a partial view of a system **100** with rod holders **160** disposed with different orientations. The system **100** as described beneficially allows for a variety of options for securing a rod holder **160** and pivot arm **180**. After the rails **140** are secured to the base plate **120**, a rod holder **160** may be secured to a pair of adjacent rails **140** in the connection channel **150**. A rod holder **160** may be secured to a pair of adjacent rails **140** by aligning the rail connection points **173** of a rod holder **160** with holder connection points **144** of adjacent rails **140** and securing by means of a mechanical connector as described herein. In some embodiments, the mechanical connector is a pin. The position of the rod holder **160** within the connection channel **150** may be changed by removing the mechanical connector securing the rod holder **160** to adjacent rails **140** and securing the rod holder **160** at a different holder connection point **144** of the rails **140**. Once the rod holder **160** has been secured to a pair of rails **140** at a desired location, the rod holder **160** is free to rotate between an angular orientation where the rod holder **160** is normal to the base plate **120** (rod holder **160a** as seen in FIG. **9**) and an angular orientation where the rod holder **160** is substantially parallel to the base plate **120** (rod holder **160b** as seen in FIG. **9**). Once the rod holder **160** has been rotated to a desirable angular orientation, the rod holder **160** may be secured in said angular orientation by aligning a pivot connection point **175** of the rod holder **160** with a holder connection point **187a**, **187b** of the pivot arm **180** and securing by means of a mechanical connector. In some embodiments, the mechanical connector is a pin. After the pivot arm **180** has been secured to the rod holder **160**, the pivot arm is secured to a pair of adjacent rails by aligning the rail connection points **191** of the pivot arm **180** with the desired holder connection points **144** of the rails **140** and securing by means of a mechanical connector. In some embodiments, the mechanical connector is a pin. The angular orientation of a rod holder **160** may be changed by removing the mechanical connector securing the pivot arm **180** to the rails **140** and securing the pivot arm **180** at a different holder connection point **144**. A holder connection point **144** that is closer to the base of the rod holder **160** will result in angular orientation that is closer to being normal to the base plate **120** than the previous angular orientation. A holder connection point **144** that is further from the base of the rod holder **160** will result in an angular orientation of the rod holder **160** that is closer to being parallel to the base plate **120** than the previous angular orientation.

[0109] The angular orientation of the rod holder may be further adjusted by removing the

mechanical connector securing the rod holder **160** to the pivot arm **180** and securing the rod holder **160** to the pivot arm **180** at a different holder connection point **187a**, **187b** of the pivot arm **180** and/or at a different pivot connection point **175** of the rod holder **160**. Adjusting the angular orientation in this manner may result in a smaller change to the angular orientation of the rod holder **160** than by changing the position where the pivot arm **180** is connected to the rails **140**. [0110] The configurations described herein are beneficial for allowing a plurality of rod holders **160** and pivot arms **180** to be attached to a plurality of rails **140** with connection channels **150** formed between a pair of adjacent rails **140** in a variety of configurations. The angular orientation and position along the length of the rails **140** of each rod holder **160** is independently configurable without altering the angular orientation and position of the other rod holders **160**.

[0111] FIG. **10** shows a cart feature **1000** that can be secured to the base plate **120**. As seen in FIG. **10** the cart feature may include at least one wheel **1010**, at least one wheel mount **1020**, a cross bar **1030** and at least one base connection feature **1040**.

[0112] The cross bar **1030** may have a cross-section that is generally rectangular or square in shape. In some embodiments, the cross bar **1030** may be cylindrical in shape. In some embodiments, the cross bar **1030** may be hollow. The cross bar **1030** may have a length that is approximately the same as a width of the base plate **120**. The wheel mounts **1020** may be connected to the cross bar **1030** at either end of the cross bar **1030**. In some embodiments, the wheel mounts **1020** may be generally L-shaped. In some embodiments, the cross bar **1030** may be connected to the wheel mounts **1020** at a bottom portion **1022** of the wheel mount **1020**, such that a portion of the wheel mount **1020** extends above the cross bar **1030**. In some embodiments, the distance between opposite wheel mounts **1020** is approximately the same as the width of the base plate **120**, such that the base plate **120** can be disposed between each of the wheel mounts **1020**. A wheel **1010** is connected to each wheel mount **1020** such that the wheel **1010** extends downward from a top portion **1024** of the wheel mount **1020**. In some embodiments, the wheels **1010** are rotatably connected to the wheel mounts **1020** such that when the cart feature **1000** is connected to the base plate **120**, the system **100** can be pushed or pulled in any direction.

[0113] The base connection features **1040** are disposed on the cross bar **1030**. There may be one or more base connection features **1040** disposed on the cross bar **1030**. In some embodiments, the base connection features **1040** may be cylindrical in shape. In some embodiments, the base connection features **1040** may be configured to be disposed through at least one large connection point **124** of the base plate **120** or through another connection point of the base plate, such as a mounting point **128** of the base plate **120**. Once the base connection features **1040** are disposed through at least one large connection point **124** or mounting point **128** of the base plate **120**, a mechanical connector, such as a pin or other connector described herein, may be placed through an upper orifice **1042** of the base connection feature **1040** to secure the cart feature **1000** to the base plate **120**. In some embodiments, each connection feature **1040** includes a rest **1044**. The rest **1044** may be square or rectangular in shape. In some embodiments, the rests **1044** may be in another shape, such as a triangle, oval, circle, polygon, hexagon or other shape. In some embodiments, the rests are configured to provide support to the base plate **120** when it is disposed on the cart feature **1000**.

[0114] In some embodiments, multiple cart features **1000** can be attached to the base plate **120** to facilitate transportation. For example, a first cart feature **1000** can be secured to a front end of the base plate **120** while a second cart feature **1000** is secured to back end of the base plate, thereby allowing the base plate **120** to be transported more easily.

[0115] FIG. **11A** shows an isometric view of a handle feature **1100**. In some embodiments, the handle feature **1100** may be utilized in conjunction with the cart feature **1000**. As seen in FIG. **11A**, the handle feature **1100** may include a handle **1110**, a central beam **1120**, and a handle base **1130**.

[0116] The handle **1110** may be connected orthogonally to the central beam **1120**, thus facilitating a user to grip and pull and/or push the handle **1110**. In some embodiments, the handle **1110** is

integrally formed with the central beam **1120**. The central beam **1120** is connected to the handle **1110**. In some embodiments, the central beam **1120** is connection to the handle at a central location. [0117] FIG. **11B** shows a bottom portion **1122** of the central beam **1120**. The central beam **1120** may have a cross-section that is substantially square or rectangular in shape with a hollow center. At a first end **1121** of the central beam **1120**, the central beam **1120** may include at least one lower base connection point **1124** and at least one upper base connection point **1126**. In some embodiments, there are two lower base connection points **1124** and two upper base connection points **1126**, a lower base connection point **1124** and an upper base connection point **1126** on a first side **1123** of the central beam **1120** and a lower base connection point **1124** and an upper base connection point **1126** on a second side **1125** of the central beam **1120**. In some embodiments, the lower base connection points **1124** are aligned along a length of central beam **1120** such that a mechanical connector can be disposed through each lower base connection point **1124** simultaneously. In some embodiments, the upper base connection points **1126** are aligned along a length of central beam **1120** such that a mechanical connector can be disposed through each upper base connection point **1126** simultaneously. In some embodiments, each of the lower base connection points **1124** and the upper base connection points **1126** may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole, a tapered hole, a keyhole slot, a dowel hole, or a helical hole.

[0118] The central beam **1120** may also include at least one protrusion gap **1128**. The protrusion gap **1128** may be located at the first end **1121** of the central beam **1120**. The protrusion gap **1128** may be of a sufficient height and width to receive the base protrusion **1160** of the handle base **1130** as described below. In some embodiments, there may be two protrusion gaps **1128**, a first protrusion gap **1128** located on a third side **1127** of the central beam **1120** and a second protrusion gap **1128** location on a fourth side (not seen) of the central beam **1120**. The first and second protrusion gaps **1128** may be aligned along a width of the third side **1127** and a width of the fourth side (not seen).

[0119] FIG. **11C** shows an isometric view of a handle base **1130**. As seen in FIG. **11C**, the handle base **1130** may include a bottom portion **1140** and a top portion **1150** and a base protrusion **1160**.

[0120] FIG. **11D** shows an isometric view of the bottom portion **1140**. The bottom portion **1140** may be in the form of an open rectangular. The bottom portion **1140** may also be hollow forming a bottom rail channel **1142**. The bottom rail channel **1142** may be of a sufficient height and width to allow a rail **140** as described above to be disposed within the bottom rail channel **1142**. The bottom portion **1140** may also include a top portion gap **1144** formed between a first end **1146** of the bottom portion **1140** and a second end **1148** of the bottom portion **1140**. The top portion gap **1144** may be of a sufficient height and width to allow a rail **140** as described above to pass through the top portion gap **1144**. In some embodiments, the top portion gap **1144** is orthogonal and/or perpendicular to the bottom rail channel **1142**.

[0121] The bottom portion **1140** may also include a top portion connection point **1147**. The top portion connection point **1147** may be centrally located on a top surface **1145** of the bottom portion **1140**. In some embodiments, the top portion connection point **1147** may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole, a tapered hole, a keyhole slot, a dowel hole, or a helical hole.

[0122] The bottom portion **1140** may also include at least one rail connection points **1149**. In some embodiments, the bottom portion **1140** includes a plurality of rail connection points **1149**. In some embodiments, the bottom portion **1140** includes a pair of aligned rail connection points **1149** on the first end **1146** of the bottom portion **1140** and a pair of aligned rail connection points **1149** on the second end **1148** the bottom portion **1140**. The bottom portion **1140** may be secured to a rail **140** as described herein by disposing the bottom portion **1140** over the rail **140** such that the rail **140** is disposed within the bottom rail channel **1142**. In this configuration, at least one rail connection point **1149** may be aligned with at least one holder connection point **144** of the rail **140**. A

mechanical connector as described herein may be disposed through the aligned at least one rail connection point **1149** and the holder connection point **144**, thereby securing the bottom portion **1140** to the rail **140**.

[0123] FIG. **11E** shows an isometric view of the top portion **1150**. The top portion **1150** may be in the form of an open rectangular. The top portion **1150** may also be hollow forming a top rail channel **1152**. The top rail channel **1152** may be of a sufficient height and width to allow a rail **140** as described herein to be disposed within the top rail channel **1152**. The top portion **1150** may also include a bottom portion gap **1154** formed between a first end **1156** of the top portion **1150** and a second end **1158** of the top portion **1150**. The bottom portion gap **1154** may be of a sufficient height and width to allow the bottom portion **1140** as described above to be disposed within the bottom portion gap **1154**. In some embodiments, the bottom portion gap **1154** is orthogonal and/or perpendicular to the top rail channel **1152**.

[0124] The top portion **1150** may be secured to the bottom portion **1140** by means of a mechanical connector connected to the top portion **1150** or integrally formed with the top portion **1150** being disposed through the top portion connection point **1147** of the bottom portion **1140**. The bottom portion **1140** may be disposed through the bottom portion gap **1154** of the top portion **1150** when secured together thereby forming a general t-shape when the handle base **1130** is assembled (as seen in FIG. **11A**).

[0125] The top portion **1150** may also include at least one rail connection points **1159**. In some embodiments, the top portion **1150** includes a plurality of rail connection points **1159**. In some embodiments, the top portion **1150** includes a pair of aligned rail connection points **1159** on the first end **1156** of the top portion **1150** and a pair of aligned rail connection points **1159** on the second end **1158** of the top portion **1150**. The top portion **1150** may be secured to a rail **140** by disposing the top portion **1150** over the rail **140** such that the rail **140** is disposed within the top rail channel **1152**. If the top portion **1150** is also secured to the bottom portion **1140**, the rail **140** is simultaneously disposed within the top portion gap **1144** of the bottom portion **1140**. In this configuration, at least one rail connection point **1159** may be aligned with a holder connection point **144** of the rail **140**. A mechanical connector may be disposed through the aligned at least one rail connection point **1159** of the top portion **1150** and the holder connection point **144** of the rail **140**, thereby securing the bottom portion **1140** to the rail **140**.

[0126] This beneficially allows a user to attach the handle base **1130** to the base plate **120** and rails **140** in a variety of configurations. For example, a user can attach the handle base **1130** to a rail **140** such that the handle base **1130** is centered along a width of the base plate **120**. In this configuration, the rail **140** may be disposed through the bottom rail channel **1142** of the bottom portion **1140**. In another configuration, the user can attach the handle base **1130** to a rail **140** such that the handle base **1130** is centered along a length of the base plate **120**. In this configuration the rail **140** may be disposed through the top rail channel **1152** of the top portion **1150** and the top portion gap **1144** of the bottom portion **1140**.

[0127] The base protrusion **1160** is connected to a top surface **1157** of the top portion **1150** of the handle base **1130**. In some embodiments, the base protrusion **1160** is integrally formed with the top portion **1150**. As seen in FIG. **11E**, the base protrusion **1160** extends in a direction normal to the top surface **1157** of top portion **1150**. In some embodiments, the base protrusion **1160** is generally parallel to the length of the top rail channel **1152**. In some embodiments, the base protrusion **1160** may be generally semi-circular in shape. In some embodiments, the base protrusion **1160** may be generally rectangular in shape. As seen in FIG. **11E**, the base protrusion **1160** may each include a central connection point **1162** and a plurality of perimeter connection points **1164**. The central connection point **1162** may be disposed centrally on the base protrusion **1160**. In some embodiments, the central connection point **1162** may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole, a tapered hole, a keyhole slot, a dowel hole, or a helical hole.

[0128] The plurality of perimeter connection points **1164** are disposed along a perimeter of the base protrusion **1160**. In some embodiments, each perimeter connection point **1164** may consist of a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole, a tapered hole, a keyhole slot, a dowel hole, or a helical hole.

[0129] The central beam **1120** may be configured to receive the base protrusion **1160** in the protrusion gap **1128**. The central beam **1120** can then be secured to the handle base **1130** by aligning the lower base connection points **1124** of the central beam **1120** with the central connection point **1162** of the base protrusion **1160** of the handle base **1130**. A mechanical connector may then be disposed through the central connection point **1162** and the lower connection points **1124** thereby securing the central beam **1120** to the base connector **350**. When the central beam **1120** is secured to the handle base **1130** by the central connection point **1162** and the lower connection points **1124**, the central beam **1120** may be free to rotate about an axis extending normally from the base protrusion **1160**. A user may then rotate the central beam **1120** to a desired orientation. Once the central beam **1120** is in a desired orientation, the central beam **1120** can be secured in the desired orientation by disposing a mechanical connector through a corresponding perimeter connection point **1164** of the base protrusion **1160** and the upper connection points **1126** of the central beam **1120**. This beneficially allows a user to change the orientation of the handle feature **1100**. In some embodiments, the orientation of the handle feature may be changed up to 180 degrees.

[0130] FIG. **12A** shows an isometric view of a t-track feature **1200**. The t-track feature **1200** includes a top surface **1210**, a bottom surface **1230** and a vertical surface **1220**. As seen in FIG. **12A**, the top surface may be generally rectangular in shape. The top surface **1210** may include at least one connection feature **1212**. The connection features **1212** are disposed on the top surface **1210**. There may be one or more connection features **1212** disposed on the top surface **1210**. In some embodiments, the connection features **1212** may be cylindrical in shape. In some embodiments, the connection features **1212** may be configured to be disposed through at least one large connection point **124** of the base plate **120** or through another connection point of the base plate, such as a mounting point **128** of the base plate **120**. Once the base connection features **1040** are disposed through at least one large connection point **124** or mounting point **128** of the base plate **120**, a mechanical connector, such as a pin or other connector described herein, may be placed through an upper orifice **1214** of the base connection feature **1212** to secure the t-track feature **1200** to the base plate **120**. In some embodiments, the connection features **1212** are removably connected to the top surface **1210**. In some embodiments, the top surface **1210** may include a connection feature point **1216**. The connection feature point may be a hole, such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole, a tapered hole, a keyhole slot, a dowel hole, or a helical hole. The connection feature point **1216** may serve to attach a connection feature **1212** to the top surface **1210**. This beneficially allows a user to change the distance between connection features **1212**.

[0131] The vertical surface **1220** may extend downward from top surface **1210**, thereby forming a t-shape with the top surface. The vertical surface **1220** may be angled such that a height of the vertical surface is smaller at a first end **1222** of the vertical surface **1220** than at a second end **1224** of the vertical surface **1220**. At the second end **1224** of the vertical surface **1220**, the vertical surface **1220** may be connected to the bottom surface **1230**. The bottom surface **1230** may be rectangular in shape. In some embodiments, the bottom surface **1230** and the vertical surface may form a t-shape (as seen in FIG. **12B**). The bottom surface may be connected to the top surface **1210** by means of a side wall **1232** extending from the bottom surface **1230** to the top surface **1210**. The bottom surface **1230** may also include a surface connection point **1234**. The surface connection point **1234** may be a hole such as, but not limited to, a round hole, a countersunk hole, a counterbore hole, a through hole, a blind hole, a tapered hole, a keyhole slot, a dowel hole, or a helical hole. The surface connection point **1234** may be configured to secure the t-track feature

1200 to an external surface or object.

[0132] In some embodiments, the t-track feature **1200** is secured to the base plate **120** by means of the connection features **1212**. The t-track feature **1200** can be slid into a corresponding t-track on an external object or surface, such as a truck, boat, car, or other vehicle, some of which are listed herein. In such a configuration, the t-shape formed by the bottom surface **1230** and the vertical surface **1220** may correspond to the t-shaped gap in the t-track. Once placed in a desired orientation, the t-track feature **1200** may then be secured to the external object or surface by means of a mechanical feature being disposed through the surface connection point **1234** and into the external object or feature.

[0133] The methods disclosed herein comprise one or more steps or actions for achieving the described method. The method steps and/or actions may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of steps or actions is required for proper operation of the method that is being described, the order and/or use of specific steps and/or actions may be modified without departing from the scope of the claims.

[0134] It should be noted that the terms “couple,” “coupling,” “coupled” or other variations of the word couple as used herein may indicate either an indirect connection or a direct connection. For example, if a first component is “coupled” to a second component, the first component may be either indirectly connected to the second component or directly connected to the second component. As used herein, the term “plurality” denotes two or more. For example, a plurality of components indicates two or more components.

[0135] Headings are included herein for reference and to aid in locating various sections. These headings are not intended to limit the scope of the concepts described with respect thereto. Such concepts may have applicability throughout the entire specification.

[0136] The previous description of the disclosed implementations is provided to enable any person skilled in the art to make or use the present disclosure. Various modifications to these implementations will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other implementations without departing from the spirit or scope of the disclosure. Thus, the present disclosure is not intended to be limited to the implementations shown herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

Claims

1. A fishing rod holding system, comprising: a base plate; a plurality of rails attached to the base plate, each of the plurality of rails comprising a plurality of connection points; at least one rod holder configured to be rotatably coupled to at least one of the plurality of rails at a first connection point of the plurality of connection points; and at least one pivot arm rotatably coupled to the at least one rod holder, the at least one pivot arm configured to be coupled to at least one of the plurality of rails at a second connection point.
2. The system of claim 1, wherein the rod holder is configured to be rotated between a first and second angular orientation when attached to the at least one of the plurality of rails.
3. The system of claim 2, wherein the first angular orientation is an orientation substantially parallel to the base plate.
4. The system of claim 2, wherein the second angular orientation is an orientation substantially normal to the base plate.
5. The system of claim 1, wherein the at least one rod holder is further configured to be detachably coupled to the at least one of the plurality of rails.
6. The system of claim 1, wherein the at least one pivot arm is further configured to secure the at least one rod holder in an angular orientation when the pivot arm is coupled to the at least one of the plurality of rails.

7. The system of claim 1, wherein the base plate is configured to be mounted onto an external surface or object.
 8. The system of claim 1, further comprising a vehicle, wherein the base plate is configured to be secured onto a surface of a vehicle.
 9. The system of claim 1, wherein the pivot arm is detachably coupled to the at least one rod holder.
 10. The system of claim 1, wherein the at least one rod holder is configured to be coupled to the least one of the plurality of rails at the first connection point by inserting a mechanical connector through a rail connection point of the rod holder and the first connection point.
 11. The system of claim 10, wherein the mechanical connectors comprise a screw, a bolt and nut, or a pin.
 12. A method for storing fishing rods or other items, the method comprising: providing a fishing rod holding system, the fishing rod holding system comprising: a base plate; a plurality of rails attached to the base plate, each of the plurality of rails comprising a plurality of connection points; at least one rod holder configured to be rotatably coupled to at least one of the plurality of rails at a first connection point; and at least one pivot arm rotatably attached to the at least one rod holder, the at least one pivot arm configured to be coupled to at least one of the plurality of rails at a second connection point; coupling the at least one rod holder to the at least one of the plurality of rails at the first connection point; coupling the pivot arm to the at least one of the plurality of rails at a second connection point; and engaging a fishing rod with the at least one rod holder, thereby securing the fishing rod to the fishing rod holding system.
 13. The method of claim 12, further comprising altering an angular orientation of the least one rod holder.
 14. The method of claim 13, wherein altering the angular orientation of the at least one rod holder comprises: detaching the pivot arm from the at least one of the plurality of rails at the second connection point; and reattaching the pivot arm to the at least one of the plurality of rails at a third connection point.
 15. The method of claim 13, wherein altering the angular orientation of the at least one rod holder comprises: detaching the at least one rod holder from the at least one of the plurality of rails at the first connection point; and reattaching the at least one rod holder to the at least one of the plurality of rails at a third connection point.
 16. The method of claim 13, wherein altering the angular orientation of the at least one rod holder comprises changing the angular orientation of the at least one rod holder between 0° degrees and 90°.
 17. The method of claim 12, further comprising mounting the base plate onto an external surface or object.
 18. The method of claim 12, further comprising securing the base plate to a surface of a vehicle.
 19. The method of claim 12, wherein coupling the at least one rod holder to the least one of the plurality of rails at the first connection point, comprises: aligning the first connection point with a rail connection point of the rod holder; and inserting a mechanical connector through the first connection point and the rail connection point.
 20. The method of claim 19, wherein the mechanical connector comprises a screw, a bolt and nut, or a pin.
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